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Telematic Intervention based on the Play Specialist Approach in the Covid-19 Era: Benefits for Parents of Children with Clinical Conditions

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Abstract*: Covid-19 pandemic has changed the routines of families all over the world. From March 2020 up to today, Italian families are still struggling for adaptation. Parents of children and adolescents with a clinical diagnosis are more at risk for parental burnout, depression, and anxiety, and they are now experiencing restrictions in many services families relied on. Home-based and hospital-based interventions based on the Play Specialist's approach have been limited due to anti-covid norms. Internationally, Play Specialist intervention has been empirically demonstrated effective in diminishing children's negative emotions in relation to medical procedures and in increasing adaptation and compliance towards medical settings. Plus, Play Specialist's intervention indirect effect on parental wellbeing is still unexplored. In Italy, differently from UK and USA, the Play Specialist intervention is not certified in the health-care system yet. The present study tests the effects on parental psychosocial health of a telematic adaptation of the Play Specialist approach (TPS), conducted in the post-lockdown months in Italy. Two groups of parents (N=33, Mean age=43.36, SD=9.81, Female= 66% receiving the TPS intervention, and N=33 Mean age=41.84, SD=6.15, Female=78% controls) of children in clinical conditions are compared. Parental burnout, anxiety, stress, depression, social support, and parental perception of children's emotional problems have been measured via self-report questionnaires. Analysis of covariance reveals that the TPS group is less stressed, perceives higher social support, lower parental burnout (i.e., emotional distancing, contrast with other/previous Self, fed-up feeling), lower emotional and behavioural child's problems than the control group. These findings are addressed at encouraging both research and practice around the Play Specialist's intervention beyond the hospital-context.

Keywords: Children's health, Covid-19, Parental Burn-out, Play Specialist.

I. INTRODUCTION

A. Covid-19 restrictive measures: the impact on families

Covid-19 pandemic has reshaped the lives of families all over the world. In Italy, mobility limitations, school and university closing, social isolation, and smart-working are impacting on family-routines since March 2020, when the Government declared the state of lockdown [1,2].

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To date – October 2020 – government measures are becoming less restrictive due to the decrease in the contagion rate in the summer months. However, the situation is still far from normal, especially for parents of children and adolescents with a clinical diagnosis who has seen restrictions in the access to many services they relied on (i.e., home-based and hospital-based assistance of Child Play Specialists) [3]. As for the majority of worldwide adult citizens, coronavirus has spread social distress, economic distress, and unemployment among parents [4,5]. A survey by the Pew Research Center shows that 91% of adults consider their life changed since the pandemic outbreak [6]. Additionally, emotions like fear and preoccupation about own's and loved ones' health and future have impacted on family wellbeing, even in families who do not have to cope with children's clinical conditions [7].

Previous research about the long-term effects of 2002-2004 SARS epidemic on the general population reports many forms of psychological distress. Specifically, depression, stress, irritability, anxiety, and PTSD symptoms are consequences of quarantine isolation [8,9]. It is worth noting that parents' and children's health conditions are strongly interlaced: according to a recent study parental stress mediates children's emotional and behavioural problems during quarantine [1]. Thus, parents exhibiting higher stress levels have to deal with children with higher behavioural and emotional problems.

One specific stress-related syndrome is parental burnout, which encompasses somatic complaints, decreases in sleep quality, perceived incompetency in the role of mother or father, and emotional distancing in the interaction with the child [10,11]. Research has pointed out several risk factors for parental burnout, such as economic insecurity, lack of leisure time, and low social support [12,13]. All of these risk factors are implied by the Covid-19 life-conditions [14]: both parents and children are at home for the majority of the time, barriers between work-time and family-time are challenging to define, and the possibility to rely on other caregiving figures (e.g., grandparents, aunts and uncles, babysitters) is limited. Parental burnout is also associated with child's maltreatment, neglect, couple conflict, and addiction, and sleep problems in parents [15]. Plus, several children's clinical conditions may facilitate parental burnout (e.g., autism, disabilities, intellectual disabilities, cancer, ADHD) as caregiving the child implies massive physical strain and psychological fatigue [16,17,18].

Nowadays, there are only a few findings about the impact of Covid-19 quarantine life-conditions in families with children and adolescents in clinical conditions. Surveys' data about caregivers of autistic children have been published, as well as operative guidelines to handle the emergency [19,20,21]. Still, a deeper investigation on how intervening to reduce the impact of anti-covid-19 restrictive measures on parents' wellbeing across different children's medical conditions (e.g., autism, disabilities, intellectual disabilities, cancer, ADHD, and other pathologies) is needed.

B. Providing the Play Specialist approach in the era of coronavirus: a telematic adaptation

Article 31 of the Convention on the Rights of the Child of the United Nations defines play as a right [22]. In this line, the European Charter of Hospitalized Children states that age-appropriate toys need to be present in the pediatric wards and hospitals. Play is an essential parameter of children's normal psychological and physical development and a source of relief during hospitalization [23]. The healing power of play is massively recognized across different cultures and societies [24]. Clinicians use play sessions to catalyze therapeutic change, helping children to face the behavioural and emotional problems connected to pathology and hospitalization [25]. According to Schaefer and Drewes, playing induces positive affects in the child, helps him/her sublimating stress, facilitates him/her to express his/her true Self (e.g., unconscious processes), encourages empathy and moral judgment, and ameliorates the attachment bond in the parent-child relationship [26]. In the nineteenth century, the founder of modern nursing F. Nightingale, stressed out the importance of playing for pediatric patients [27]. Frauman and Gilman documents that F. Erikson was the first nurse to systematically using play sessions, discovering its advantages in preparing children for medical procedures [28]. Thus, the Child Play Specialist or Child Life Specialist is a facilitator for children who have to cope with illness and adapt to the hospital context [29]. In the hospital, this figure works with the pediatric team by developing a personalized play program for each child and adolescent [30,31]. Play Specialists sessions can encompass different scenarios: i. medical play to familiarize with medical procedures or equipment [32]; ii. playing with puppets, dolls, marionettes, or art and music to encourage the child to express and communicate [33]; iii. playing with pets to mitigate perceived pain and to increase positive emotions [34,35]; iv. digital games with educational and socialization purposes [36]. The Play Specialist intervention's positive effects on children, across-different medical conditions, and countries, have been widely demonstrated. Playing sessions significantly reduce pediatric patients' anxiety and stress levels (e.g., before surgery) and ameliorate coping strategies [37,38,39,40,41]. On the other hand, the indirect effects of the Child Play Specialist intervention on parents are still ignored. It is also important to note that the medical setting is not the only context in which Play Specialists can operate [42]. Play Specialist's interventions are also implemented within palliative home-care, with autistic children who benefit from a reassuring and predictable context, and with other pathological conditions [42,43,44]. In Italy, the corona virus outbreak negatively impacted on the possibility to provide hospitalized children with the Play Specialists' approach. Hospitals limited access to the pediatric department, excluding Play Specialists who come from external institutions. In fact, differently from UK and USA, the Italian Health Ministry has not encompassed this figure yet in the national healthcare system: the regional administration of Liguria's district is evaluating to certificate the Play Specialist's training promoted by the Porto dei Piccoli charity. Thus, the term used to describe the intervention in the present research is Play Specialist, and not Child Play Specialist (certified in the UK) or Child Play Specialist (certified in the USA), although the theoretical and methodological background are similar. Covid-19 restrictions have led to adapt the Play Specialist approach for the online environment. In the light of a recent meta-analysis supporting the beneficial effects of a wide range of web-based health interventions in pediatric patients [45], this crisis has represented both a challenge and an opportunity to reshape the Play Specialist approach in the online environment. Specifically, the present study intends to verify whether and how a telematic adaptation of Play Specialist's intervention (TPS) beneficially impacts on families by focusing on variables defining parental wellbeing (i.e., parental burnout, anxiety, stress, depression, perceived social support).

C. Study Aims and Hypotheses

The present study aims at shedding light on the effect of a Telematic Play Specialist-based (TPS) intervention over parents and children. In particular, the following hypotheses have been tested: (a) TPS approach decreases parental-burnout, stress, anxiety, and depression; (b) TPS approach increases parental perceived social-support; (c) TPS approach decreases children's emotional and behavioural problems.

II. METHOD

A. Sample

The sample included N=66 parents (Mean age=42.61, SD=8.15, F=72%). Parents have been recruited in August and September 2020. Parents have been divided in two conditions: (a) TPS group consisting of parents from families with children currently participating to the TPS intervention provided by Porto dei Piccoli charity (Genoa, Italy) through two-times a week video-call sessions (i.e., on Skype, Zoom, Meet, Teams); (b) Control group consisting of parents from families with children not currently participating to the TPS (e.g., waiting list). TPS group includes N=33 parents (Mean age=43.36, SD=9.81, F= 66%). Control group includes N=33 parents (Mean age=41.84, SD=6.15, F=78%). All the participants have been asked for informed consent about data collection and provided with information on the ethical standards followed (i.e., Italian psychologists deontological code, Italian Psychology Association ethical code, Italian law 56/98, and declaration of Helsinki).

TPS intervention group includes parents of N=33 children with mean age=11.09 (SD=4.01) with the following diagnoses: ADHD (9%), Autism Spectrum Disorders (18%), Cytomegalovirus (3%), Intellectual Disability (15%), muscular dystrophy (3%), encephalomyelitis (3%), hydrocephaly (3%), hypoacusis (6%),

genetic disease (9%), renal pathologies (6%), pluri-malformation (3%), tetra paresis (18%), plus non-specified conditions.

The control group includes parents of N=33 children with mean age=8.55 (SD=4.77) with the following diagnoses: ADHD (9%), Autism Spectrum Disorders (9%), Bardet-Biedl Syndrome (3%), Brugada Syndrome (3%), Down Syndrome (6%), cerebral palsy (3%), Cancer (3%), Williams Syndrome (60%), plus non-specified conditions.

B. Measures

Self-report questionnaires have been collected to measure the variables of interest.

Socioanagraphic and general information have been collected to control the effect of: parent's gender, parent's age, child's age, parent's occupation, parent's educational level, family socioeconomic status (SES), number of sons/daughters, family type, and other ongoing treatment.

Parental Burnout Assessment (PBA) has been used to measure parental burnout [46] through ad-hoc translation. PBA consists of 23 phrases describing states (e.g., emotions, cognitions, feelings) of the parent-child relationship (e.g. "*I feel I am no more capable of showing my children how much I love them*"). Parents' had to answer on a Likert-scale how frequently they are experiencing each state from 1=never to 7=every day. PBA allows to measure four dimensions of parental burnout: Emotional Distancing (EU) in the interactions with the child, Exhaustion (EX) related to the parental role, Conflict with Other (Previous) Self (CO) (e.g., perceiving themselves as a worse parent than before), Fed Up feeling (FU) (e.g., not enjoying the parental role anymore).

Multidimensional Scale of Perceived Social Support (MSPSS) has been administered to measure how the parent perceives him/herself supported by his/her social framework [47,48]. MSPSS consists of 12 affirmations about social relationships (e.g., "*I have someone in my life who really cares about my feelings*") to measure perceived social support. Answers are ordered on a Liker-scale of agreement (from 1= I totally disagree to 7= I totally agree). MSPSS allows detecting three dimensions of Social Support: from Significant Others, Family Support, and Friends' Support.

Depression Anxiety and Stress Scale-21 (DASS-21) has been administered to measure parental psychological health [49,50]. It consists of 21 phrases describing the emotional states felt in the last 7 days (e.g., "*I was not capable of feeling positive emotions*"). Answers are rated on a Likert-scale of frequency (from 0= It has never happened to me to 3= it has always happened to me). The scale allows detecting three scores: Depression, Anxiety, and Stress.

Cantril scale has been administered to measure quality of life as a controlling variable (covariate) [51]. Parents have been showed the picture of a stair where the lowest step (0) represents living the worst life possible, and the highest step (10) matches with the best life possible. Participants had to provide a number from 0 to 10, indicating which step corresponded to their life.

Finally, Strengths and Difficulties Questionnaire (SDQ) has been used to measure how parents perceived their children from a behavioural and emotional perspective [52,53]. SDQ consists of 25 assertions describing the child's behaviours and emotional states in the last month from the parent's perspective (e.g., "*He/She (son/daughter) often seems worried*"). Answers are ordered on a Likert-scale of agreement (from 1=not true, 2= partially true, 3= true). SDQ consents to detect the following dimensions of child emotions and behaviours: Conduct Problems, Emotional Problems, Problems with Peers, Hyperactivity, Prosociality, Internalizing problems, Externalizing Problems.

C. Analytic Plan

The present study has used Analysis of Covariance to verify hypotheses a,b,c, by comparing the two groups of parents (experimental and control group). Covariates have been inserted to control the effects of: parent's age, nationality, parent's gender, child's age, parent occupation (dichotomous: having a job vs not having a job), parental education (e.g., middle school, high school, academic degree, post-academic degree), the parent's life satisfaction measured with Cantril scale, socioeconomic status, the number of sons/daughters, family type (e.g., traditional, single-parent family, stepfamily), treatment (dichotomous: the family members have experienced psychotherapy, psychiatric intervention, counseling, coaching). The analyses have been conducted on JASP 0.11.1.

III. RESULTS

Analyses of covariance has been implemented to compare TPS group and control group. Several covariates' significant effects have emerged in between ANCOVAs only for the following variables: (a) parent's stress: socioeconomic status $F(1,55)=8.52$ $p=.01$ partial $\eta^2=.13$, number of sons $F(1,55)=8.11$ $p=.01$ partial $\eta^2=.13$, family type $F(1,55)=6.06$ $p=.02$ partial $\eta^2=.11$; (b) anxiety: socioeconomic status $F(1,55)=5.17$ $p=.02$ partial $\eta^2=.08$, (c) depression: socioeconomic status $F(1,55)=5.12$ $p=.03$ partial $\eta^2=.08$; (d) hyperactivity: child's age $F(1,55)=4.65$ $p=.03$ partial $\eta^2=.07$; (e) child's problem with peers: parent's age $F(1,55)=5.66$ $p=.02$ partial $\eta^2=.09$; (f) child's prosociality: number of sons/daughters $F(1,55)=7.47$ $p=.01$ partial $\eta^2=.12$; (g) externalization: child's age $F(1,55)=4.87$ $p=.03$ partial $\eta^2=.08$, parent's educational level $F(1,55)=4.75$ $p=.03$ partial $\eta^2=.08$; (h) significant other's support: family type $F(1,55)=6.28$ $p=.01$ partial $\eta^2=.10$.

ANCOVAs results are summarized in Table-I. ANCOVA models reveal a significant group effect in stress $F(1,55)=55.69$ $p<.001$ partial $\eta^2=.51$, perceived social support $F(1,55)=4.27$ $p=.04$ partial $\eta^2=.07$, parental burnout $F(1,55)=4.84$ $p=.03$ partial $\eta^2=.08$, emotional distancing (ED) $F(1,55)=4.62$ $p=.03$ partial $\eta^2=.07$, contrast with other/previous self (CO) $F(1,55)=6.17$ $p=.01$ partial $\eta^2=.10$, fed-up feeling (FU) $F(1,55)=5.92$ $p=.01$ partial $\eta^2=.09$, perceived emotional $F(1,55)=7.76$ $p=.007$ partial $\eta^2=.12$ and conduct child's problems $F(1,55)=5.41$ $p=.02$ partial $\eta^2=.09$.

Table- I: ANCOVA results
[INSERT TABLE-I HERE]

[a.] Note: Degrees of Freedom= 1,55. SDQ=Strengths and Difficulties Questionnaire, ED=Emotional Distancing, FU=Fed up, CO=Contrast with Other Self, EX=exhaustion. Statistical significance at *=p<.05, **p<.01, ***p<.001.

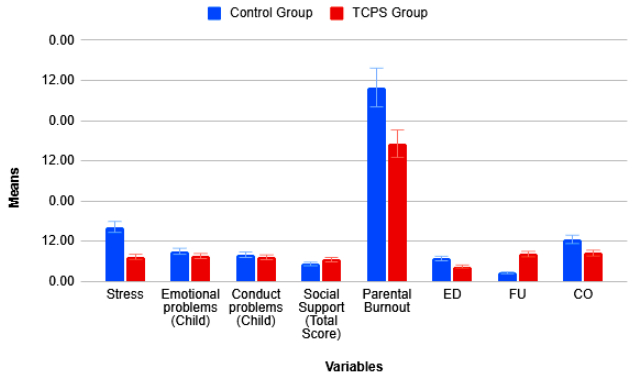

Fig.1 Means of Control Group and TPS Group.

IV. DISCUSSION

The present research examines whether parents of children in a wide range of medical conditions benefitted from TPS intervention in the months after Italian covid-19 lockdown [2]. Parents of children participating to the TPS refer to be less stressed and score lower than controls in total parental burnout and several sub-dimensions (i.e., emotional distancing with child, contrast with other/previous parental Self, feeling fed up with parental role). Moreover, the TPS group perceives higher social support and lower children's emotional and behavioural problems.

Given that covid-19 post-lockdown months may have constituted a risk factor for Italian parents' health and may increase parental burnout in vulnerable families, these findings respond to the scientific debate over the positive effects of web-based interventions to support pediatric patients [12,13,14,15,16,17,18,45]. These data align with recent research demonstrating the association between parents' and children's wellbeing during covid-19 outbreak in Italy [1]. In such an interlaced model of family health, this is one of the first studies examining the Play Specialist approach's indirect effects of parents, in addition to previous evidence on Play Specialist approach's positive outcomes on children [37,38,39,40,41]. Plus, these findings demonstrate that the Play Specialist approach can be practiced online with positive outcomes beyond home-based and hospital-based settings, and for a broader range of pediatric patients (e.g., ADHD, cytomegalovirus, intellectual disability, muscular dystrophy, encephalomyelitis, hydrocephaly, hypoacusis, genetic disease, renal pathologies, pluri-malformation, tetra paresis, and other non-specified conditions) than Autism Spectrum Disorders.

It is worth noticing that three covariates resulted statistically significant in association with parental stress: (i.) low socioeconomic status, considering that socioeconomic insecurity may constitute a consequence of covid-19 pandemic; (ii.) number of sons/daughters which increases parental efforts; (iii.) family structure because being a single parent implies to deal with parental responsibilities and fatigue alone.

The present research has methodological limitations. The first regards the use of self-report questionnaires: these instruments may have impacted the results [54]. Social biases (e.g., social desirability) and compilation biases (e.g., the attraction of scale's extreme points) may have distorted parental responses. Plus, the version of PBA is an ad-hoc translation which has not been validated in Italy yet with factor analysis. Thus, future studies may integrate data collection with qualitative interviews. Second, implementing a cross-sectional design does not provide information on the intervention over time. This may suggest the importance to compare groups longitudinally by developing, in the future, mixed research designs. Third, ANCOVA models do not reveal the TPS effect on anxiety, depression, other child's problems (e.g., hyperactivity, problems with peers, prosociality levels, externalization, internalization), and parental exhaustion. Further studies may consider expanding the sample size to further verify inter-group comparisons for these variables.

V. CONCLUSION

The group of parents whose son/daughter participated to TPS results less stressed, perceives higher social support, lower parental burnout (e.g. emotional distancing, contrast with other/previous Self, fed-up), lower emotional and behavioural (i.e., conduct) child's problems than the parents of the control group. These findings suggest to a wide range of professionals (e.g., pediatricians, psychologists, nurses, social workers) the importance of the Play Specialists approach beyond the hospital or home-based setting. These results also highlight the positive effects of telematic intervention of the intervention based on the Play Specialist's approach on families of children in clinical conditions in the months after Covid-19 lockdown in Italy.

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REFERENCES

1. M. Spinelli, F. Lionetti, M. Pastore, & M. Fasolo, "Parents' Stress and Children's Psychological Problems in Families Facing the COVID-19 Outbreak in Italy", *Frontiers in Psychology*, vol. 11, 2020, pp. 1713.
2. Government, "*Italian Government: Measures to Face the Coronavirus Covid-19*", March 2020, Available: http://www.governo.it/it/coronavirus
3. J. Lee, "Mental health effects of school closures during COVID-19", *The Lancet Child & Adolescent Health*, vol. 4 n. 6, 2020, pp. 421.
4. E. McGinty, R. Presskreischer, H. Han, & C. Barry, "Psychological Distress and Loneliness Reported by US Adults in 2018 and April 2020", *Journal of the Medical Association JAMA*, vol. 324, n. 1, 2020, pp. 93-94.
5. A. Kirzinger, L. Hamel, C. Muñana, A. Kearney, & M. Brodie, "*KFF Health Tracking Poll-Late April 2020: Coronavirus, Social Distancing, and Contact Tracing*", Kaiser Family Foundation. Available: http://www.kff.org/report-section/kff-health-tracking-poll-late-april-2020-economic-and-mental-health-impacts-of-coronavirus/
6. Pew Research Center, "*Most Americans Say Coronavirus Outbreak Has Impacted Their Lives*", 2020. Available: htTPS://www.pewsocialtrends.org/wp-content/uploads/sites/3/2020/03/PSDT_03.30.20_W64-COVID-19.Personal-impact-FULL-REPORT.pdf
7. J.J. Liu, Y. Bao, X. Huang, J. Shi, & L. Lu, L., "Mental health considerations for children quarantined because of COVID-19", *The Lancet Child & Adolescent Health*, vol. 4, 2020, pp. 347–349.

8. X. Liu, M. Kakade, C. Fuller, B. Fan, Y. Fang, J. Kong, & P., Wu, "Depression after exposure to stressful events: lessons learned from the severe acute respiratory syndrome epidemic", *Comprehensive Psychiatry*, vol. 53, n. 1, 2012, pp. 15-23.
9. L. Hawryluck, W. Gold, S. Robinson, S. Pogorski, S. Galea, & R. Styra, "SARS control and psychological effects of quarantine, Toronto, Canada." *Emerging infectious diseases*, vol. 10, n. 7, 2004, pp. 1206.
10. R.R. Abiden, "Parenting Stress Index: a measure of the parent–child system", in *Evaluating stress: A book of resources*, C.P. Zalaquett, R.J. Wood, Ed. Lanham, MD: Scarecrow Press, 1997, pp. 277-291.
11. I. Roskam, M.E. Raes, & M. Mikolajczak, "Exhausted parents: development and preliminary validation of the parental burnout inventory", *Frontiers in psychology*, vol. 8, 2018, pp. 163.
12. C. Lindström, J. Åman, & A. L. Norberg, "Parental burnout in relation to sociodemographic, psychosocial and personality factors as well as disease duration and glycaemic control in children with Type 1 diabetes mellitus", *Acta Paediatrica*, vol. 100, n. 7, 2011, pp. 1011-1017.
13. M. Sorkkila, & K. Aunola, "Risk factors for parental burnout among Finnish parents: The role of socially prescribed perfectionism", *Journal of Child and Family Studies*, vol. 29, n. 3, 2020, pp. 648-659.
14. A. K. Griffith, "Parental burnout and child maltreatment during the COVID-19 pandemic", *Journal of Family Violence*, 23 June 2020, pp. 1-7.
15. M. Mikolajczak, M. E. Brianda, H. Avalosse, & I. Roskam, I., "Consequences of parental burnout: Its specific effect on child neglect and violence", Child Abuse & Neglect, vol. 80, 2018, pp. 134-145.
16. M. J. Weiss, "Hardiness and social support as predictors of stress in mothers of typical children, children with autism, and children with mental retardation", *Autism*, vol. 6, n. 1, 2002, pp. 115-130.
17. N. Beheshtipour, P. Nasirpour, S. Yektatalab, M. Karimi, & N., Zare, "The effect of educational-spiritual intervention on the burnout of the parents of school age children with cancer: A randomized controlled clinical trial", *International Journal of Community Based Nursing and Midwifery*, vol. 4, n. 1, 2016, pp. 90.
18. S. M. Arafa, & M. Lamlom. Burnout Syndrome in Caregivers of Children with ADHD. *Arab Journal of Psychiatry*, vol. 31, n, 1, 2020, pp. 54-68.
19. M. Carly Albaum, P. Modica, J. Gorter, R. N. Nazilla Khanlou, V. Lee, & J. Weiss, J. "*Impact of COVID-19 on the Mental Health and Wellbeing of Caregivers and Families of Autistic people*", June 2020. Available: htTPS://irsc-cihr.gc.ca/
20. M. Colizzi, E. Sironi, F. Antonini, M. Ciceri, C. Bovo, & L. Zoccante, "Psychosocial and Behavioral Impact of COVID-19 in Autism Spectrum Disorder: An Online Parent Survey", *Brain Sciences*, vol. 10, n. 6, 2020, pp. 341.
21. A. Narzisi, "Handle the autism spectrum condition during Coronavirus (COVID-19) stay at home period: Ten tips for helping parents and caregivers of young children", Brain Sciences, 2020, vol. 10, n. 4, pp.207.
22. L. Lundy "Children's rights and educational policy in Europe: the implementation of the United Nations Convention on the Rights of the Child", *Oxford Review of Education*, vol. 38 n. 4, 2012, pp. 393-411.
23. K. Koukourikos, L. Tzeha, P. Pantelidou, & A. Tsaloglidou, "The importance of play during hospitalization of children", *Materia Socio-medica*, vol. 27, n. 6, 2015, pp. 438.
24. B. Sutton-Smith, "Evolving a consilience of play definitions: Playfully". In *Play and culture studies*, S. Reifel, Ed, New York: HarperCollins, 1999, pp. 239-256.
25. A. A. Drewes, "Play-based interventions", *Journal of Early Childhood and Infant Psychology*, vol. 2, 2006, pp.139–156.
26. C. E. Schaefer, & A. A. Drewes, "The therapeutic powers of play and play therapy", In *Blending play therapy with cognitive behavioral therapy: Evidence-based and other effective treatments and techniques*, Ed., New York, NY: Walley, 2009, pp. 3-15.
27. A. Francischinelli, F. Almeida, & D. Fernandes, "Routine use of therapeutic play in the care of hospitalized children: nurses' perceptions". *Acta Paulista de Enfermagem*, vol. 25, n. 1, 2012, pp. 18-23.
28. A. C. Frauman, & C. M. Gilman, "Creating a therapeutic environment in a pediatric renal unit", *ANNA journal*, vol. 16, n. 1, 1989, pp. 20-2.
29. A. Webster, "The facilitating role of the play specialist", *Paediatric Nursing*, vol. 12, n. 7, 2000, pp. 24.
30. A. Dolan, "A day in the life of a hospital play specialist", *The British Journal of Theatre Nursing: Natnews: the Official Journal of the National Association of Theatre Nurses*, vol.3, n. 3, 1993, pp. 31-32.
31. P. Hart-Spencer, & A. Griffiths, "How Child Play Specialists can reduce use of anaesthesia during radiotherapy", *Cancer Nursing Practice*, vol. 14, n.10, 2015, pp. 16-20.
32. K. McCue, "Medical play: An expanded perspective", *Children's Health Care*, vol. 16, n. 3, 1988, pp. 157-161.
33. A.M. Sposito, F. Montigny, V.Sparapani, R. Lima RA, F. M. Silva-Rodrigues et al., "Puppets as a strategy for communication with Brazilian children with cancer", *Nursing & Health Sciences*, vol. 18, n. 1, 2016, pp. 30-37.
34. M. Kaminski, T. Pellino, J. Wish, "Play and pets: the physical and emotional impact of child-life and pet therapy on hospitalized children", *Children's Health Care*, vol. 31 n. 4, 2002, pp. 321-335.
35. L. Vagnoli, S. Caprilli, C. Vernucci, S. Zagni, F. Mugnai & A. Messeri, A. "Can presence of a dog reduce pain and distress in children during venipuncture?", *Pain Management Nursing*, vol. 16 n. 2, 2015, pp. 89-95.
36. S. Jurdi, J. Montaner, F. Garcia-Sanjuan, J. Jaen, & V. Nacher, "A systematic review of game technologies for pediatric patients", *Computers in Biology and Medicine*, vol. 1, n. 97, 2018, pp. 89-112.
37. C. Gill, "Helping children cope with renal disease: the role of play specialist". *Journal of Renal Nursing*, vol. 2, n. 5, 2010, pp. 244-247.
38. E. R. Moore, K. L. Bennett, M. S. Dietrich, & N. Wells, "The effect of directed medical play on young children's pain and distress during burn wound care", *Journal of Pediatric Health Care*, vol. 29, n. 3, 2015, pp. 265-273.
39. W. Li, J. Chung, K. Ho, & B. Kwok, B. "Play interventions to reduce anxiety and negative emotions in hospitalized children", *BMC Pediatrics*, vol. 16, n. 1, 2020, pp. 36.
40. C. L. Wong, W. Y. Ip, B. Kwok, K. Choi, B. Ng, & C. W. H. Chan, "Effects of therapeutic play on children undergoing cast-removal procedures: a randomised controlled trial", *BMJ open*, vol. 8, n. 7, 2018, pp. e021071.
41. A. M. Ullan, & M.H. Belver, "Integrative Pediatrics and Child Care Play as a Source of Psychological Well-Being for Hospitalized Children: Study Review", *Integrative Pediatrics and Child Care*, vol. 2, n. 1, 2019, pp. 92-98.
42. C. Hubbuck, "*Play for sick children: Child Play Specialists in hospitals and beyond*", London: Jessica Kingsley Publishers.
43. A. Arnaert, & M. Wainwright, "Providing care and sharing expertise: Reflections of nurse-specialists in palliative home care", *Palliative & Supportive Care*, vol. 7, n. 3, pp. 357.
44. O. T. Sari, "Outcomes of play-based home support for children with autism spectrum disorder", *Social Behavior and Personality: an international journal*, vol. 42 n. 1, 2014, pp. 65S-80S.
45. A. Rantala, M. Pikkarainen, J. Miettunen, H. G. He, & T. Pölkki, T., "The effectiveness of web-based mobile health interventions in paediatric outpatient surgery: A systematic review and meta-analysis of randomized controlled trials", *Journal of Advanced Nursing*, vol. 76, 2020, pp. 1949–1960.
46. I. Roskam, M. E. Brianda, & M. Mikolajczak, "A step forward in the conceptualization and measurement of parental burnout: The Parental Burnout Assessment (PBA)", *Frontiers in Psychology*, vol. 9, 2018, pp. 758.
47. G. D. Zimet, N. W. Dahlem, S. G. Zimet, & G. K. Farley, "The multidimensional scale of perceived social support". *Journal of Personality Assessment*, vol. 52 n. 1, 1988, pp. 30-41.
48. M. Prezza, & M. C. Principato, "The social web and Social Support" in *Conoscere la comunità*, M. Santinello Ed. Bologna, Italy: Il Mulino, 2002, pp. 193-233.
49. J.D. Henry, & J. R. Crawford, "The short-form version of the Depression Anxiety Stress Scales (DASS-21): Construct validity and normative data in a large non-clinical sample", *British Journal of Clinical Psychology*, vol. 44, n. 2, 2005, pp. 227-239.
50. G. Bottesi, M. Ghisi, G. Altoè, E. Conforti, G. Melli, & C. Sica, "The Italian version of the Depression Anxiety Stress Scales-21: Factor structure and psychometric properties on community and clinical samples", *Comprehensive Psychiatry*, vol. 60, 2015, pp. 170-181.
51. H. Cantril, "*The pattern of human concerns*". New Brunswick, NJ: Rutgers University Press, 1965.
52. R. Goodman, "*Youthinmind. SDQ: Information for Researchers and Professionals about the Strengths & Difficulties Questionnaires*", 1997, London: Youthinmind.
53. V. Tobia, & G. M. Marzocchi, "The strengths and difficulties questionnaire-parents for Italian school-aged children: psychometric properties and norms", *Child Psychiatry & Human Development*, vol. 49, n. 1, 2018, pp. 1-8.
54. C. Dicken. "Good impression, social desirability, and acquiescence as suppressor variables", *Educational and Psychological Measurement*, vol. 23, 1963, pp. 699–720.

AUTHORS PROFILE

Giulia Perasso obtained a M.Sc degree in Psychology in 2015 (University of Pavia, Italy) with first honor. In 2016 she researched as visiting intern at Middlesex University (London, UK), at the Centre for Abuse and Trauma Studies. In 2017, she obtained a M.Sc degree in Criminology at Sapienza University (Rome, Italy) and she joined the national Register of Psychologists. In 2018 she became intervener of VIPP-SD (Videofeedback to Promote Positive Parenting and Sensitive Discipline) achieving the certification by University of Leiden (NL). She is now a PhD Candidate in Psychology, Neuroscience and Data Science at University of Pavia (Italy) with a thesis on risk-behaviours of children aging from 11 to 15. Giulia authored scientific papers published in international journals (European Journal of Developmental Psychology, and the Journal of Family Studies, Health and Addiction), and Italian journals (Italian Journal of Psychology, Clinical and Developmental Psychology, Psychological Research). She also wrote chapters for Zeigler Hill's "Encyclopedia of Personality and Individual Differences", and for Silton's handbook "Exploring Best Child Development Practices in Contemporary Society". She has conducted referring for the journal Psychological Reports. Her main interests are developmental psychology and developmental psychopathology. Nowadays she is collaborating as consultant researcher at the Porto dei Piccoli Onlus (Genoa, Italy) where she studies the beneficial effects of the approach based on Child Play Specialist's intervention on families of children with pathological conditions (ASD, intellectual disabilities, ADHD, behavioural problems, genetic syndromes, cancer) or undergoing surgical procedures.

Maggiore Alice obtained a M.Sc degree in Clinical and Community Psychology in 2017 (University of Genoa, Italy), and joined the Professional Register of Psychologists in 2019. In 2017 she took part to a research by University of Genoa in the field of SLD (Specific Learning Disorders), examining the correlation between motivation to study in adolescents and academic performance through the enhancement and empowerment of self-image and self-esteem towards one's own abilities. In 2018 she worked at La Spezia (Italy) penitentiary as a Psychologist and Educator, collaborating with the team of territorial services (Ser.T), social services, cooperatives, and rehabilitation communities, mainly working to support inmates in detoxification and autonomy. She also worked together with the team of educators and psychiatrists chosen by the Director of the penitentiary, in order to help the prisoners in reintegrating into society, mediating between the penitentiary and potential employers, supporting the inmates' autonomy and reintegration. Since 2019 she has been working with Porto dei Piccoli (Genoa, Italy), as educator, following the Child Play Specialist approach, operating across-Italy in many services and hospitals and working with children and families with fragile conditions/backgrounds or pathologies (psychiatric disorders, genetic diseases / hereditary, neurodegenerative diseases, oncohematological diseases, surgical operations); nowadays she is collaborating in experimental research to study the benefits of the telematic intervention of the Child Play Specialist on the wellbeing of the families.

Chiara Allegri is a Psychologist and an operator of Porto dei Piccoli charity (Genoa, Italy). In 2012 she obtained a B.sc in Personality and Relational Psychology at University of Padua (Italy) with a thesis around adolescents with hypoacusis experiences of social network sites. In 2015, she obtained a M.Sc degree in Work and Communication Psychology at University of Padua (Italy) with first honor. Her M.Sc thesis consisted of a cross-cultural study of emotion recognition (e.g., in Italy and Portugal), developed under the supervision of Professor A. Mass. Chiara Allegri joined the national Psychologists Register in 2018. In 2019, she achieved a M.Sc in Psychology of Emergency. During 2016, she worked at TAMAT ONG (Perugia, Italy) were she gained expertise in designing European social projects, by taking part to Social Stat Ups (Erasmus + KA2). Since 2017, Chiara is a Porto dei Piccoli (Genoa, Italy) operator, working according to the Child Play Specialist approach with hospitalized and inpatient children. In particular, she supports children with clinical conditions in many pediatric departments of hospitals in northern and center Italy (e.g., neuropsychiatry, maxillofacial, odontolaryngologist, etc.). Nowadays, she has achieved the role of Porto dei Piccoli Coordinator, implying human resources management, and cross-district national coordination of Porto dei Piccoli projects (i.e., Genoa, Turin, Mantua, Rome).

Gloria Camurati has twenty years of experience in school teaching with children and adolescents. She is GM (General Manager) of the charity Porto dei Piccoli (Genoa, Italy). She founded Porto dei Piccoli in 2005, aiming at creating a network of services to support families of children in clinical conditions. When she founded Porto dei Piccoli, Gloria Camurati addressed at combining her philanthropy with her love for the sea. In fact, the main seat of Porto dei Piccoli is Genoa, one of the main ports of the Mediterranean area. The term Porto dei Piccoli is translated as "Children's Harbor", metaphorically evocating socio-emotional safety as in Bowlby's Attachment Theory. Plus, the association is supported by shipping companies on the territory. Porto dei Piccoli works in synergy with northern Italy hospitals to provide families with interventions based on Play Specialists' approach into the pediatric wards, at home, and through web-based platforms. Porto dei Piccoli also implements school-based projects to decrease stereotypes over illness among children and interactive experiences (e.g., trips) to get children in clinical conditions to familiarize with the sea. Gloria Camurati is also a pioneer in establishing a Play Specialists' training in Italy, drawing local administration's attention on this figure. She is a member of Porto dei Piccoli scientific and consultive committee and she is part of the Editorial team of Porto dei Piccoli quarterly magazine. She is directing research on Porto dei Piccoli projects to corroborate the charity reputation through scientific findings and international partnerships.

Table I- ANCOVA results

	Group Mean		Analysis of (Co)Variance			
	Control (N=33)	*TPS* (N=33)	*SS*	*MS*	*df*	*F*
Stress	15.78(3.61)	7.24(4.1)	574.71	574.71	1	55.69***
Anxiety	3.63(3.76)	2.97(5.36)	6.21	6.21	1	0.51
Depression	5.91(5.36)	4.51(4.56)	24.76	24.76	1	1.03
SDQ Total Score	46.93(5.31)	44.33(6.19)	98.88	98.88	1	3.04
Emotional problems (Child)	9.03(2.37)	6.97(2.55)	45.19	45.19	1	7.76**
Conduct problems (Child)	8.01(1.52)	7.15(1.21)	9.88	9.88	1	5.41*
Hyperactivity (Child)	9.48(1.43)	10.01(2.11)	0.03	0.03	1	0.01
Peer problems (Child)	9.66(1.38)	9.39(1.73)	3.82	3.81	1	1.63
Prosociality (Child)	10.75(2.41)	10.81(3.24)	2.83	2.83	1	0.36
Externalization (Child)	17.48(2.53)	17.15(2.92)	8.69	8.69	1	1.24
Internalization (Child)	18.69(3.39)	16.36(2.71)	75.31	75.31	1	7.92
Social Support (Total Score)	5.16(1.22)	5.93(0.98)	5.15	5.15	1	4.27*
Significant Others	5.52(1.49)	6.19(1.21)	4.12	4.12	1	2.32
Family Support	5.29(1.51)	6.03(1.22)	7.31	7.31	1	3.74
Friends Support	4.65(1.42)	5.57(1.05)	4.31	4.31	1	2.71
Parental Burnout	57.54(33.42)	40.66(13.84)	3158.56	3158.56	1	4.84*
ED	6.54(3.51)	4.31(1.42)	36.15	36.15	1	4.62*
FU	1.97(8.01)	8.12(3.08)	220.31	220.31	1	5.92*
CO	12.36(7.96)	8.31(2.71)	221.28	221.28	1	6.17*
EX	26.66(15.65)	19.93(8.81)	433.04	433.04	1	2.78